

LESSON 7

KARNAUGH MAPPING



- In 1953, Maurice Karnaugh was a telecommunications engineer at Bell Labs.
- While exploring the new field of digital logic and its application to the design of telephone circuits, he invented a graphical way of visualizing and then simplifying Boolean expressions.
- This graphical representation, now known as a Karnaugh map, or Kmap, is named in his honor.

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- A Kmap is a matrix consisting of rows and columns that represent the output values of a Boolean function.
 - The output values placed in each cell are derived from the minterms of a Boolean function.
 - A *minterm* is a product term that contains all of the function's variables exactly once, either complemented or not complemented.

Rules for K-Maps

- 1. Each cell with a 1 must be include in at least one group.
- 2. Try to form the largest possible groups.
- 3. Try to end up with as few groups as possible.
- 4. The number of 1s in a group must be a power of 2 – even if it contains a single 1.
- 5. Groups can be formed only at right angles; diagonal groups are not allowed.
- 6. Groups can overlap and wrap around the sides of the Kmap.

Rules for K-Maps

7. The larger a group is, the more redundant inputs there are:

- A group of 1 has no redundant inputs.
- A group of 2 has 1 redundant input.
- A group of 4 has 2 redundant input.
- A group of 8 has 3 redundant input.
- A group of 16 has 4 redundant input.

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Two Variable K-Map

A	B	F
0	0	a
0	1	b
1	0	c
1	1	d

Truth Table.

A \ B	0	1
0	a	b
1	c	d

F.

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Three Variable K-Map

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A B		0	1	3	2
		4	5	7	6



A	B	C	D	X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
1	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Four Variable K-Map

A \ B				
	0	1	3	2
0	4	5	7	6
12	13	15	14	
8	9	11	10	

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Don't Care Condition

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	x
0	1	1	1
1	0	0	1
1	0	1	x
1	1	0	0
1	1	1	0

A \ B	0	1	3	2
	4	5	7	6